

Development of Smart Elderly Care Mobile Application for Health Management System

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Abstract— Elderly care management has great potential to be further developed and shaped with smart healthcare solutions. In general, elderly care is defined as a service that serves the needs and requirements of senior citizens. As age progresses, the physical strength, health condition and mental stability deteriorate, especially when people reached a certain age (also known as elderly). It also brings about more medical appointments, medication and health issues. For elderlies, it is not easy to have a systematic management on medical appointments, medications and their health status on their own. It would be difficult for them to update or explain in details on their health conditions to their children or authorized caregivers. In most cases, family members are taking turns and responsibility to take their parents for medical appointments and check-up. It is quite a challenging task to keep up with the health status, medication details and others if more than one person act as the caregivers. This project proposes a smart health management system called Smart Elderly Care App (SECA). The main objective of this project is to develop a mobile apps to facilitate the health management issues. In general, SECA has five main features which are medical appointment, medication, daily health data, health summary, and elderly profile. By having these features, users can shape elderly care with innovative healthcare solutions to help the elderly people and their caregivers in the health management.

Keywords—Elderly care, smart health application, health management.

I. INTRODUCTION

The elderly population in Malaysia is defined as people aged 65 and above. From 2018 to 2019, the elderly population grew from 2.12 million to 2.21 million in less than a year. The population of Malaysians aged 65 and above is estimated to reach 7% in 2020. It is expected that the population will grow to 15% in 2040 [1]. It is vital to provide better care for the elderly population as it is increasing by the day. It becomes one of the escalating challenges to society and healthcare systems in overcoming the crisis of illness. For the theme of "Shaping Malaysia for Smart Healthcare Solutions," elderly care has its great

potential to be further developed and shaped with intelligent healthcare solutions.

As people are aging, it comes with increasing health risks and conditions [2]. One may have few health conditions and need regular medical appointments. They may require someone to take them to attend regular check-ups and to monitor their health status from time to time. In general, several issues arise when caring for the elderly. In this case, children or family members of the elderly may not always be available to take care of them due to various factors. As a result, they will seek assistance from others, such as authorized personal caretakers or elderly care facilities.

Elderly care issues demand a lot of responsibilities, including keeping track of various medical appointments, recording multiple medications and dosages, monitoring long-term health conditions, tracking daily health parameters, documenting elderlies' health profiles, and many more. In most cases, the record keeping process is performed manually which is prone to mistakes. The development of health management services involving mobile phones has been increasing and important to assist the monitoring process of health status of individual in their daily lives [3, 4]. However, the elderly may experience specific challenges in adapting to constantly evolving technology [5].

In most cases, elderly people are being taken care by a group of authorized caregivers, for example their children taking their turns to take their parent for medical appointments at the same time. In this situation, an ineffective hand-over from one caregiver to the next might results in missing medical appointments and medications, especially if the care givers are engaged with other commitments. Next, each individual may have a separate set of medications to take and a schedule of medical visits to keep. In this case, it would be difficult for the care givers to keep on track of everything without any mistakes.

Additionally, it is difficult for family members to get a complete overview of the elderly's health condition. In this case, family members have to record most of the information by themselves, rather than using a common

platform that allow them to access all medical information of the elderly in order to keep track of the elderly's health condition. Subsequently, the elderly might not be able to remember all of their health details, especially the updated additional health details that are usually given after each medical appointment. In other words, older adults are finding it harder to keep track of their health status.

As life expectancy increases, it is imperative that the elderly take advantage of the benefits of technology to remain active and independent. Mobile health applications and real time health monitoring are widely used nowadays as they promote a healthy lifestyle and self-management of diseases, opening new horizons in the interactive health service delivery [6, 7, 8, 9]. In addition to everyday activities, the use of mobile apps has impacted the way the population has managed healthcare. There are applications to support medication administration to monitor blood pressure, to schedule medical appointments, among others. The aim to assist the elderly in healthcare has been one of the objectives for developers to develop a website or an app that could help the elderly in terms of health. Most of the developed apps offers limited main features.

For example, the Pill Monitor [10] designed for iPhone users which reminds the elderly on medication schedule. This friendly reminder medication app is created specifically for the registered nurses who are taking care of the elderly. Another example is the BookDoc [11]. Users could easily access the app to generate a medical appointment. Users could search and make a booking session with their preferred healthcare professionals anytime at anywhere. However, the app is still unstable based on different users' feedback from Google Play. For example, users can only generate their medical appointment at private clinics or hospitals.

In this paper, a simple and user-friendly mobile application for the health management system called Smart Elderly Care App (SECA) is developed. The specifications are set as follows:

- Simple and easy to use
- Offers a few important healthcare features
- Contains safe, robust, and stable online database
- Low cost of development, utilization, and maintenance

II. METHODOLOGY

This section describes the methodology to conduct this project.

A. Online Survey

In order to understand the needs of the health management system or mobile apps for the elderly people, an online survey was conducted. Based on the survey analysis, the important features of the mobile apps are identified. A quick online survey was conducted to understand the eldercare health management issues which involve elderlies, caregivers, and others. The goal of this survey is to identify the healthcare related problems encountered by the respondents while taking care of elderly. The results of this survey is shown in Figure 1. There were 91 respondents; which are, 77 (85%) are female while 14 (15%) are male. Majority of the respondents fall in the age range of 40 to 49

years old -46 respondents (51%), followed by 20 respondents (22%) from age range of 50 to 59 years old, and 10 respondents (11%) from age range of 20 to 29 years old. The main objective of this survey is to identify important features of the mobile apps.

Based on the online survey, a healthcare application is mainly needed so that it can help users to organize their health management in a systematic way. Figure 1 shows the survey results of preferred features to be integrated to the proposed mobile apps. Based on the survey outcomes, the top four preferred features are, reminder on medical appointments, management of medical appointments, keeping track of elderly medication and finally store the elderly's medical history.

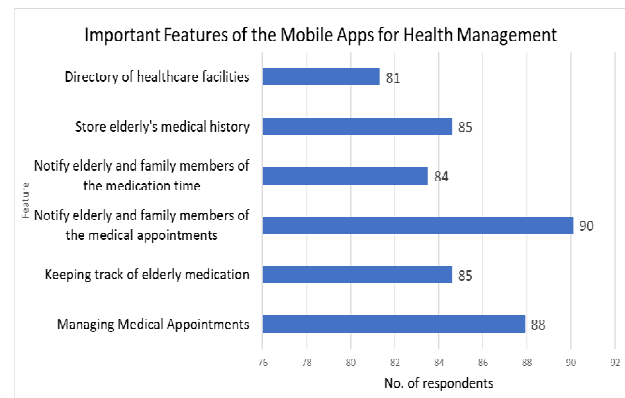


Fig. 1. Important features of the Mobile Apps from the Online Survey

B. Software Development

In this project, SECA utilizes Firebase as the online database. In fact, Firebase is safe because it can limit the amount of people that can access all users' data. To be more particular, Firebase can only allow authorized persons to access all users' data. For non-authorized persons, they can only access their own data in SECA, but not in Firebase. Next, Firebase is a robust database. In cases where there is any problem with the mobile phone, it will not affect the data that are stored in Firebase. In other words, the data in Firebase can survive from SECA failures due to the great robustness of Firebase.

SECA is constructed using Android Studio, which is a free programming platform that requires zero cost of development. For the utilization cost, users can download and install SECA for free from different mobile stores, especially Google Play Store. Apart from that, the usages of different SECA features are totally free.

C. System Overview

The system block diagram of SECA is as shown in Figure 2. Based on Figure 2, users can gain access to SECA that is developed using Android Studio and Firebase. In order to use SECA features, users have to login into their account. If they are new users, they can login after creating their account. If users forgot their password, they can login after resetting their password. After logging in, it indicates that users have gained their full access to utilize SECA features.

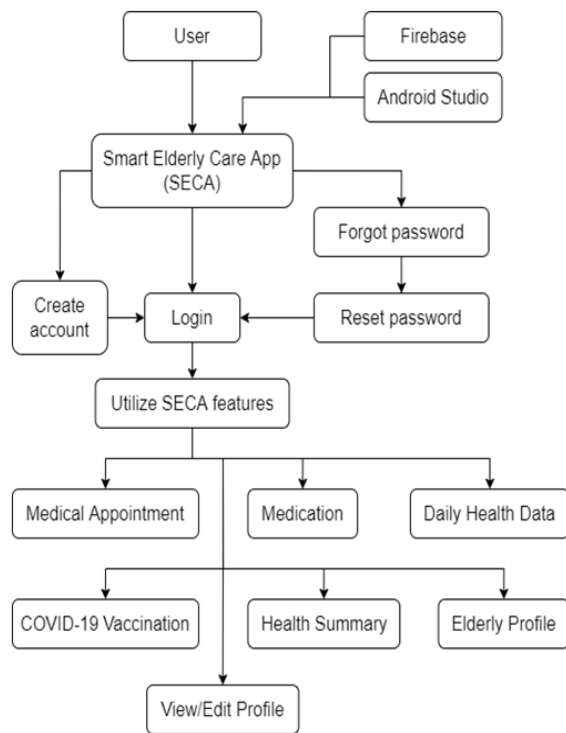


Fig. 2. System block diagram of SECA

Basically, SECA has five main features, which are Medical Appointment, Medication, Daily Health Data, Health Summary, and Elderly Profile. For Medical Appointment section, users can generate different medical appointments which help to remind them about their upcoming schedules. In Medication section, user is able to key in different types of medicines so that they can easily show their medication intake to any medical expert for quick recognition of certain medicines. For Daily Health Data section, users may record their health information on daily basis. In Health Summary section, users can insert some of their previous health histories, such as appointment reports or surgical reports. Finally, the fifth main feature is the profile, where they can edit their elderly profile whenever they want to update about their latest health summary. Additionally, SECA has one extra important feature relevant to the current condition, which is the Covid-19 vaccination record. Finally, SECA has one common feature that allows users to view or edit their personal profile. In this case, this personal profile is totally different compared to Elderly Profile. For this personal profile, it stores all personal information about users.

User can utilize five main features, one extra feature, and one common feature. Specifically, the five main features are referring to Medical Appointment, Medication Record, Daily Health Data, Health Summary, and Elderly Profile. The flowchart for each feature is demonstrated in Fig. 3-7. The extra and common feature are COVID-19 Vaccination and View/Edit Profile, respectively. After executing any feature, users can continue to use SECA by executing other features. If they wish to stop, users can end the usage of SECA.

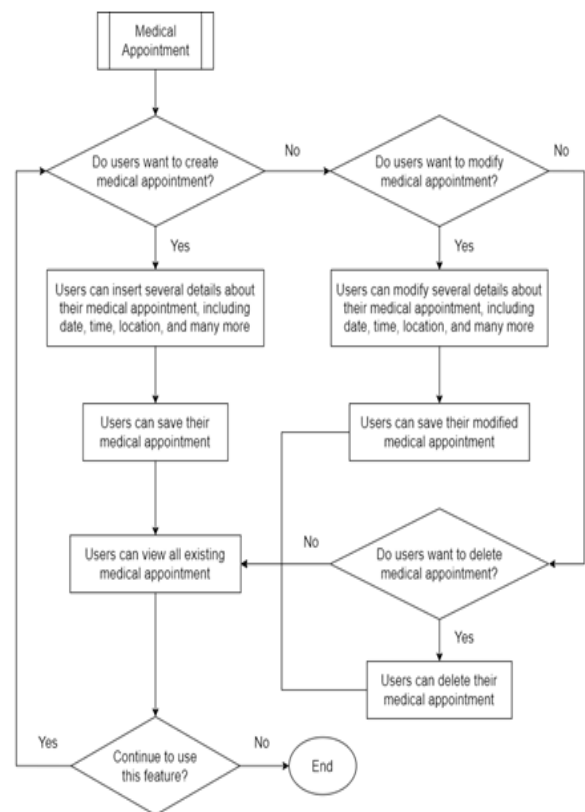


Fig. 3. Flowchart of Medical Appointment

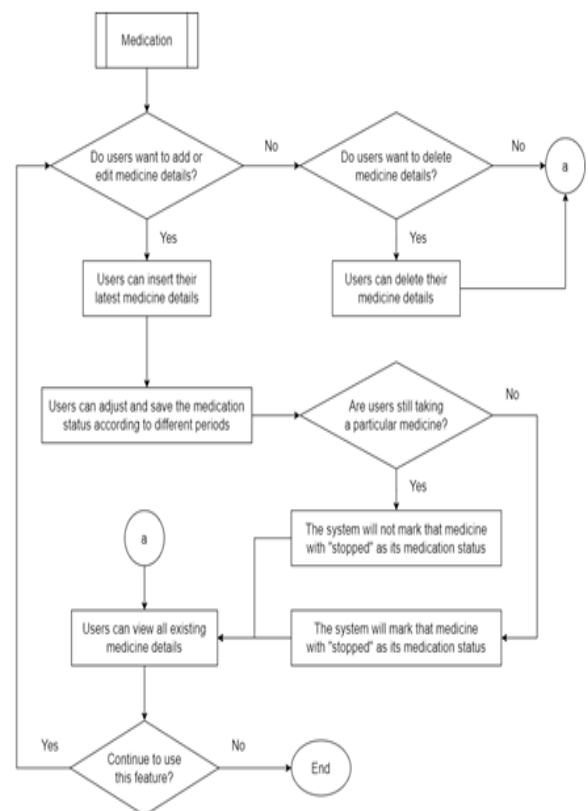


Fig. 4. Flowchart of Medication Record

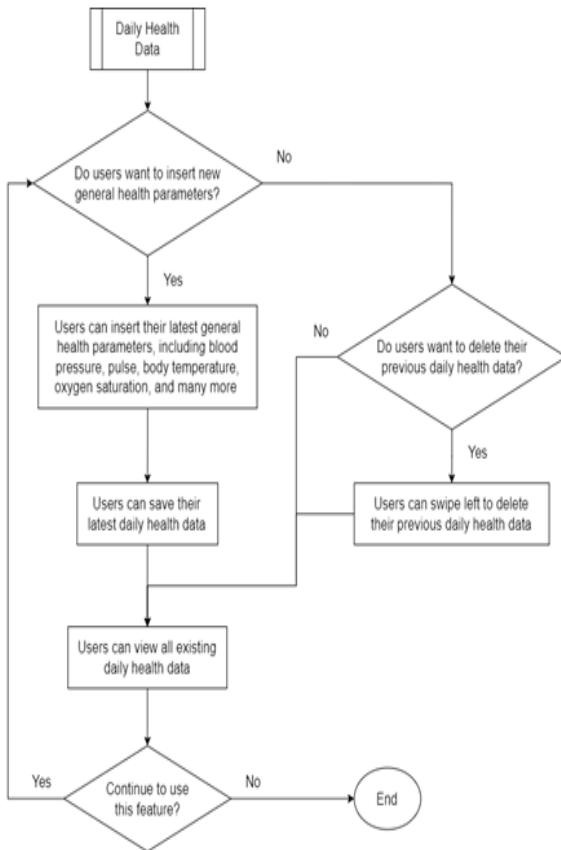


Fig. 5. Flowchart of Daily Health Data

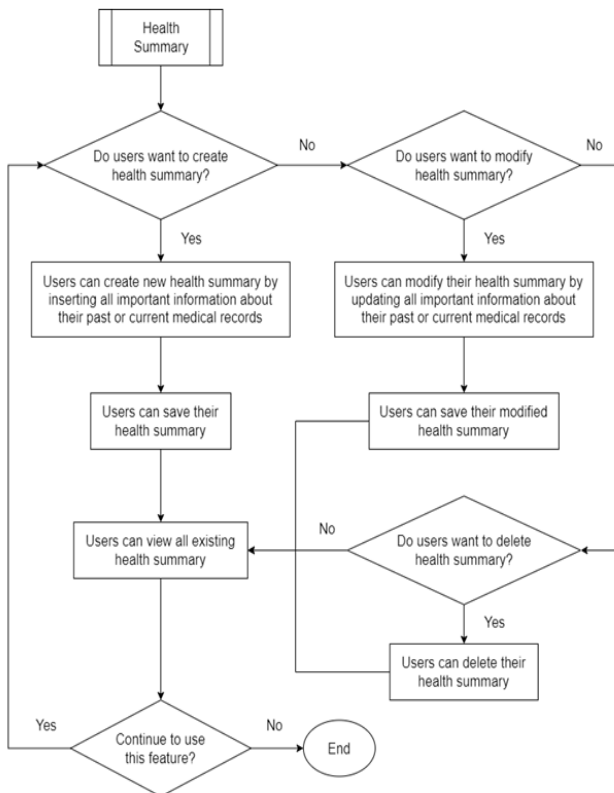


Fig. 6. Flowchart of Health Summary

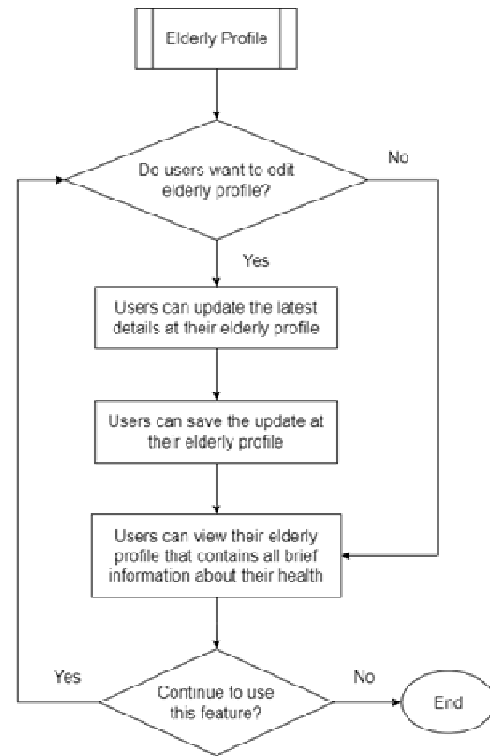


Fig. 7. Flowchart of Elderly Profile

III. RESULTS AND DISCUSSION

Based on the described features, the mobile application of SECA is developed in a user-friendly interface. The login page is set as the launcher activity of the application. In other words, users will be directed to this page when they launch the application. For new users, they can tap the 'Register Now' button to register a new account. To login into SECA, existing users can enter their email address and password at the spaces provided, then, tap the 'Login' button. The email address and password entered will be crosschecked with the database. Users will be able to log in to their respective account if the details entered matched with the database. Otherwise, an error message will be shown.

Figure 8 shows the main page of SECA. The profile photo, username, and NRIC of user will be displayed at the main page. User will be able to view or edit their SECA profile by tapping 'View/Edit Profile'. The other buttons are to buttons to access the features of SECA which are, 'Medical Appointment', 'Medication', 'Daily Health Data', 'COVID-19 Vaccination', 'Health Summary', and 'Elderly Profile'. User will be directed to the features by tapping on the respective button. Finally, user can log out of their account by tapping the 'Logout' button.

Figure 9(a) and 9(b) shows the menu interface of the Medical Appointment and Medication, respectively. The Medical Appointment feature is designed to assist users in managing their medical appointment. Users can create a new appointment by tapping 'Create an appointment' button. They will then be directed to the 'New Appointment' page. In addition, it can be also edited to implement any changes.

The second feature of SECA is Medication. The interface is shown as in Figure 9(b). This feature is designed to assist users in managing their medication. Some of the users might have several types of medication to be consumed in a day.

The dosage, times of consumption per day and instruction of consuming (e.g., before food, after food, with food etc.) may differs for different medication. The list of medications added will be displayed at this page, showing their respective photo and details. They are sorted by the name of medication in ascending order to ease users' reference. Users can add a new medication by tapping 'Add new medication' button. They will then be directed to the 'New Medication' page.

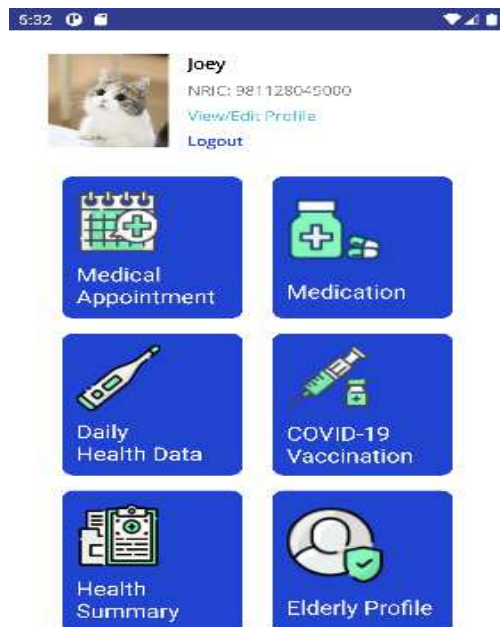


Fig. 8. Main Page of SECA Mobile Apps

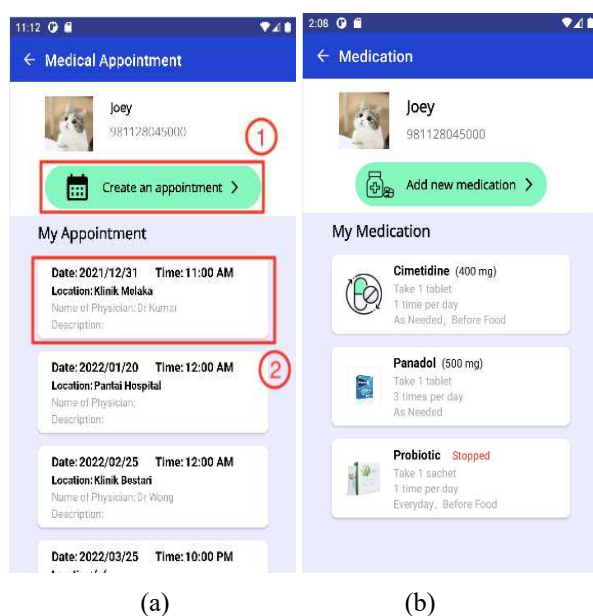


Fig. 9. Medical Appointment and Medication Page

Figure 10 demonstrates the Daily Health Data and Health Summary. The daily health data (as in Figure 10(a)) is designed to assist users in monitoring their health parameters on a daily basis. The parameters include blood pressure, heart rate, temperature, oxygen saturation level (SpO2), and

blood glucose level. It might be difficult to keep track of elderly's health parameters accurately without the aid of technology. Hence, this feature aims to monitor the daily health parameters of elderly, so that early detection of some disease can be triggered. The health entries added will be displayed at this page. They are sorted by the time of entry in descending order, from the most recent to the least recent. The height and weight of the user will be shown at the top of the page. By tapping the 'Log new health data' button, users will be directed to the 'New Health Data' page.

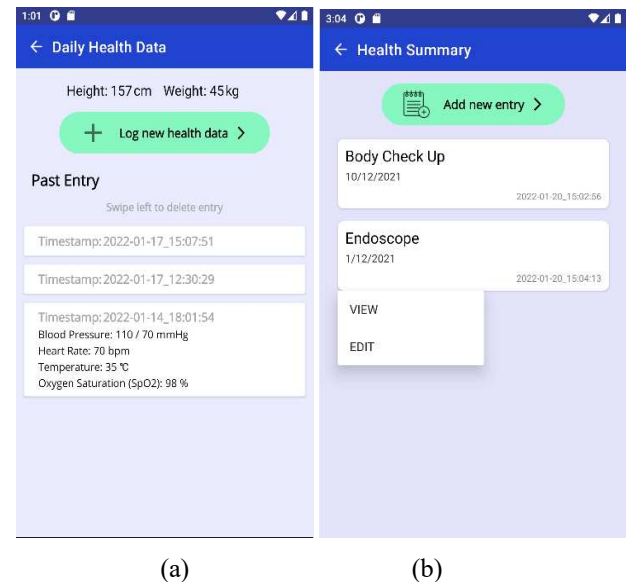


Fig. 10. Daily Health Data & Health Summary Page

The Health Summary (as in Figure 10(b)) is designed to assist users in recording their health summary. It might be difficult to keep track of elderly's health record accurately without the aid of technology. Hence, this feature aims to keep the health records of elderly. The health entries added will be displayed at this page. They are sorted by the title of entry in ascending order to ease users' reference.

The fifth is the Elderly Profile (refer Figure 11). This feature is designed to store health information of elderly. The information include gender, blood type, allergies, and disabilities of users. An extra space is provided for users to add comment too. Also, an emergency contact section is also designed to store basic information of the emergency contact person of elderly in the case of any emergency.

The data of the application is stored in a database named Firebase. The basic information of users, data of appointment and daily health data are stored in the Cloud Firestore. Most apps need to know the identity of a user. Knowing a user's identity allows an app to securely save user data in the cloud and provide the same personalized experience across all of the user's devices. Firebase Authentication provides backend services, easy-to-use SDKs, and ready-made UI libraries to authenticate users to your app. It supports authentication using passwords, phone numbers, popular federated identity providers like Google, Facebook and Twitter, and more.

The development of this software has the potential to be widely used in elderly care facilities and healthcare facilities. Elderly who are staying with their caretakers or family members can also utilize this application as their healthcare management assistant tool. Relatives of elderly

will be able to monitor the health condition as well as get updates on medication and appointment of elderly even from afar. A few improvements are possible to be made to suit the demand and need of the end users to be more user friendly and more convenient. The media uploading feature could be designed for video uploading and multiple images uploading in 'Medication' and 'Health Summary'. Also, SECA could be designed such that users are able to change to language of the application, so that we can accommodate a more diverse user base. Other than that, a healthcare facilities directory feature could be added to the application. This will help the users to locate the hospitals, clinics, or specialist nearby.

Elderly Health Profile	
	Joey Poh 981128045000 01110821782
Address	Parkland Condominium, 75000 Melaka.
Date of Birth	28/11/1998
Age	24
Gender	Female
Blood Type	B+
Allergies	Dairy
Disabilities	None
Comment	AF experience age 20
Emergency Contact	
Name	Catherine
Phone No.	0123456789
Address	Parkland Condominium, 75000 Melaka.
Relationship	Mother

Fig. 11. Elderly Profile

IV. CONCLUSION

A mobile application to assist the health management, SECA (Smart Elderly Care Application) is successfully developed. It comprises five main features, which are Medical Appointment, Medication, Daily Health Data, Health Summary, and Elderly Profile. With these useful features, SECA can help to manage the medical appointments and the medications of the elderly. Apart from

that, SECA can help to keep track of daily health parameters of the elderly. Additionally, SECA can provide a complete overview of the elderly's health records which will ease the handover process of caretakers.

ACKNOWLEDGMENT

The authors would like to extend appreciation to School Electrical and Electronic Engineering, Universiti Sains Malaysia and to all correspondents of the conducted survey.

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